

- 32** The partial pressure of oxygen in blood vessels in different parts of the body affects the percentage saturation of each haemoglobin molecule with oxygen.

blood vessel	partial pressure of oxygen / kPa	saturation of each haemoglobin molecule with oxygen / %
vein in active muscle	4	25
vein in resting muscle	7	75
arteriole in lungs	12	100

Which statement explains the difference in saturation between a resting and an active muscle?

- A** A small decrease in partial pressure causes two atoms of oxygen to leave each haemoglobin molecule because the oxygen dissociation curve is sigmoid shaped.
- B** A small decrease in partial pressure between 4 kPa and 7 kPa causes two molecules of oxygen to leave each haemoglobin molecule because the oxygen dissociation curve is steep.
- C** Loading of oxygen is slower between 4 kPa and 7 kPa due to binding of one oxygen molecule to each haemoglobin molecule making the next one harder to bind.
- D** Loading of oxygen is faster between 4 kPa and 7 kPa due to binding of one oxygen molecule to each haemoglobin molecule making the next one easier to bind.